

Introduction

Leptomeningeal disease (LMD) is a severe and often terminal complication of advanced cancers, characterized by the infiltration of malignant cells into the leptomeninges and dissemination throughout the cerebrospinal fluid (CSF). While LMD is most commonly associated with breast cancer, it is particularly challenging in patients with estrogen receptor-positive (ER+) subtypes, accounting for up to 15% of metastatic breast cancer cases. The prognosis for LMD remains poor, with limited treatment options and median survival typically less than six months. A major clinical obstacle in ER+ breast cancer is the development of resistance to endocrine therapies such as tamoxifen and imlunestrant (LY3484356). This resistance limits long-term therapeutic effectiveness and contributes to disease progression, including metastasis to the leptomeninges. Imlunestrant is a novel, orally bioavailable selective estrogen receptor degrader (SERD) currently under clinical investigation. Unlike earlier SERDs, imlunestrant demonstrates robust ER α degradation and brain penetrance, making it a promising candidate for treating LMD in endocrine-resistant settings. To address the lack of preclinical models for therapy-resistant LMD, this study aims to establish an in vivo model using tamoxifen-resistant MCF7-luciferase cells. This model enables the evaluation of imlunestrant's therapeutic efficacy and provides a platform to explore new treatment strategies for this intractable disease.

Conclusion

This project establishes the foundational steps for modeling leptomeningeal disease in tamoxifen-resistant ER+ breast cancer using intracranial injection of luciferase-labeled MCF7 cells. Our findings highlight the cisterna magna as the most effective injection route for inducing consistent and visible dissemination within the cerebrospinal fluid, confirmed through Evans Blue dye testing. By optimizing delivery techniques and implementing estrogen support to mimic hormonal conditions necessary for tumor viability, we have completed a critical phase of model development. This preclinical framework sets the stage for evaluating imlunestrant therapeutic potential over the next 5–8 weeks. If successful, this approach could contribute to the development of more targeted and brain-penetrant treatments for leptomeningeal metastases in endocrine-resistant breast cancer—a clinical context with currently limited options and fire prognosis.

Materials and Methods

Cell Culture:

Tamoxifen-resistant MCF7 (MCF7/TamR) breast cancer cells expressing luciferase were cultured in 5% CO₂. Cells were cultured in DMEM containing Tamoxifen and expanded for in vivo use.

Animal Model:

Mice were injected with either 2% Evans blue or 0.4% Methylene blue prior to cell delivery in the cisterna magna to confirm location. Evans blue proved to be superior due to its strong binding affinity and deeper tissue penetration which increased visibility in the models. 10 female immunodeficient SCID mice were anesthetized and inserted with estrogen pellets a day before intraventricular injections. To induce leptomeningeal disease (LMD), 20,000 luciferase-tagged MCF7/TamR cells were delivered directly into the **cisterna magna** or the **lateral ventricle** using a stereotaxic frame (coordinates: 0.5 mm posterior and 1.1 mm lateral to bregma, depth 2.5 mm).

Drug Treatment:

Upon detection of tumor signal by IVIS, mice will be randomized into two groups (n=5/group). One group will receive **implunestrant (10 mg/kg)** via **oral gavage**, once daily, 3 days per week, while the control group remained untreated.

Monitoring:

Tumor burden was tracked twice weekly using **IVIS bioluminescence imaging**, along with body weight measurements and basic neurological assessments. At study completion (5–8 weeks), mice will be euthanized, CSF will be harvested, and **histological analysis of brain and spinal cord** will confirm LMD and treatment efficacy.

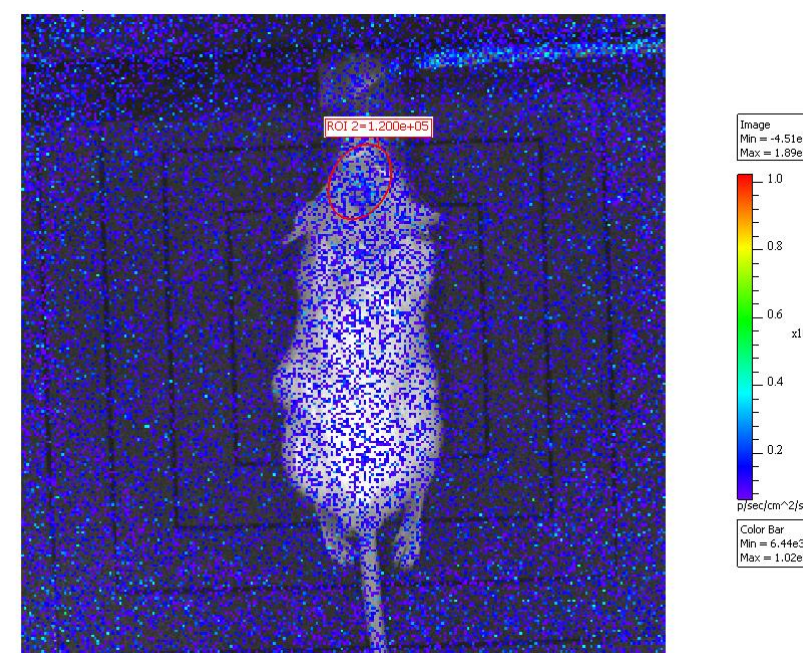


Figure 1. IVIS imaging showing bioluminescence and tumor signal in trial mouse

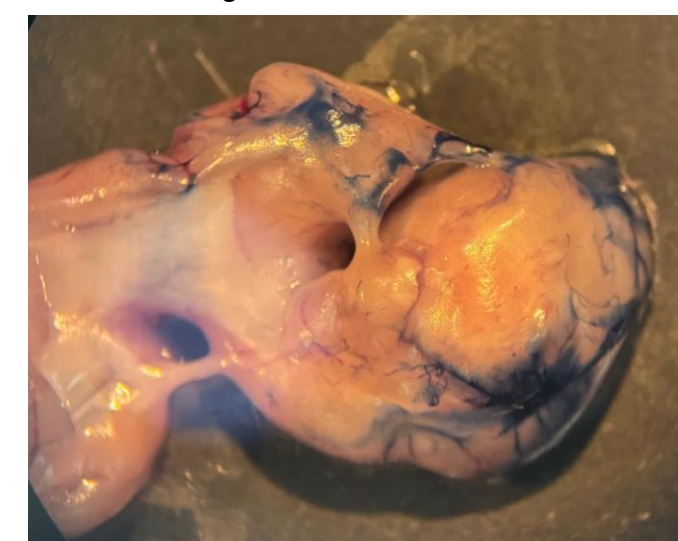


Figure 3. Evans Blue dye distribution following cisterna magna injection.

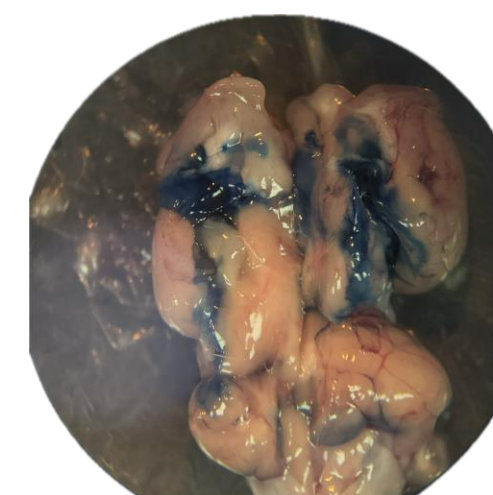


Figure 2. Evans Blue dye visibility after stereotaxic lateral ventricle injection at 6 mm depth.

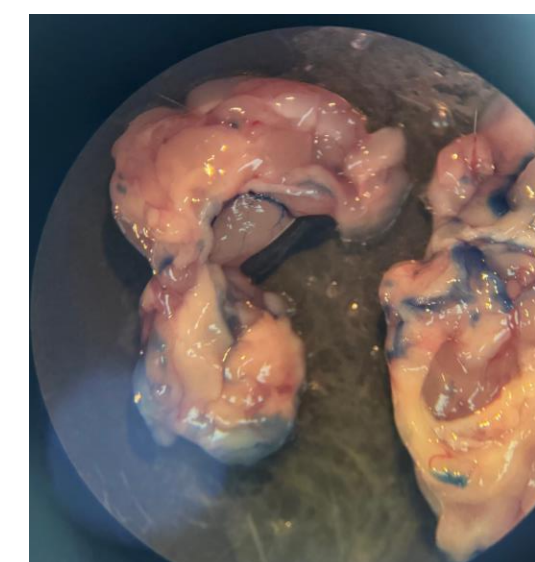


Figure 4. Evans Blue dye visibility after stereotaxic lateral ventricle injection at 4 mm depth.

Results

1. Dye Comparison:

- 2% Evans Blue provided significantly better visibility than Methylene Blue due to its binding to albumin and high CSF permeability.
- 0.4% Methylene Blue quickly diffused or faded, limiting its utility for visual confirmation.

2. Injection Route Optimization:

- Two delivery methods tested:
- Stereotaxic injection into the lateral ventricle at depths of 4 mm, 5 mm, and 6 mm
- Cisterna magna injection via surgical exposure behind the brainstem
- 6 mm depth showed the greatest visibility of Evans Blue in the stereotaxic group, but still inferior to cisterna magna delivery.
- Cisterna magna injections produced the most reliable and robust dye distribution, confirming it as the preferred route for cancer cell delivery.

3. Current Status:

- Estrogen pellets have been implanted to support tumor engraftment and drug action (critical due to ESR1 mutation sensitivity).
- Finalized protocol for MCF7-luciferase cell delivery and treatment workflow.
- Work successfully completed: model development and procedural optimization.

References

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